SEBASTIAN MONZÓN

Melrose, MA 02176 | +1 (781) 530-8016 | smonzon360@gmail.com | **WWW:** https://smonzon.com

RESEARCH INTERESTS

Multimodal Interaction, Generative AI, Computational Sensing & Design

EDUCATION

05/2023

B.S. Computer Science, Minor in Business
University of Massachusetts Amherst (Honors College) — Amherst, MA

- Thesis Title: *Towards Generative Adversarial Audio Synthesis via Latent Interpolation Between Desired Timbral Qualities*
- GPA: 3.68/4.0

AWARDS

NSF Fellowship (CSGrad4US)

- **$159,000** over 3 years for a 5-year PhD fellowship period (Starting Fall 2026).

Scholarships

- Melvin Howard Scholarship, Ruth M. White Scholarship, UMass Amherst Dean's Award

HackUMass IX Hackathon Awards

- Won a finalist award and 'Best use of Google Cloud' for a Spotify song queue voting app.

CONFERENCES & PRESENTATIONS

Symphony of Dewatering Water Harvesting Hydrogels using AI-Generated Music
2025 Material Research Society Fall Meeting, Boston (12/2025)

- Presenting a novel generative AI framework for improving hydrogel water extraction with music.

Music Generation with Stable Diffusion Inpainting
Dept. of Multifunctional Metamaterials (META) at MIT (04/2025)

- Presented a generative AI framework for music generation using Stable Diffusion/inpainting to Dr. Boriskina's cohort.

PUBLICATIONS IN PREPARATION

An AI-Driven Acoustic Symphony: Optimized Hydrogel Dewatering Leveraging a Diffusion-model for Music Spectrogram Inpainting
Shuvo, I., Monzón, S., Smith, G., Flores, R., Diaz-Marin, C., Gu, R., Nagle, S., Paradiso, J., Boriskin, V., Boriskina, S.

- Designed and tested an AI framework for improved hydrogel air-water extraction using music generated via Mel-spectrogram Stable Diffusion (Riffusion).
- Analyzed hydrogel response to the GTZAN music dataset to inversely design optimal waveforms with optimal frequency ranges.
- Experimented with soft prompting, kernel ridge, SHAP, and other ML methods.

A Wearable Waterproof System for Long-Range Health Monitoring and Intelligent Distress Signaling
Shuvo, I., Monzón, S., Boriskina, S.

- Designed, 3D printed, and coded an embedded swim-cap device with long-range radio (ESP32 w/ LoRa), IMU, and pulse sensor for real-time machine learning interpretation of swimmer movement and biomedical data.
- Programmed to transmit, display, and record movement, heart rate, and heart rate variability (HRV) data in real-time with a LoRa receiver.
- Tested underwater for waterproofing and packet loss.

Musically Interpreting a Simple Generative Transformer
Cosme-Clifford, N., Symons, J., Kapoor, K., Monzón, S., Wm. White, C.

- Explored the aptitudes and failures of a GPT-style model with self-attention, trained on choral-style hymn MIDI files.
- Analyzed the learned latent representations for musical recurrence and chord probabilities.
- Investigated existing self-attention methods for music transformers.

ADDITIONAL RESEARCH EXPERIENCE

MAGE: Motion-to-Audio Generative autoEncoder *(06/2025 - Present)*

A multi-modal variational auto-encoder (MMVAE) for hand-gesture to audio synthesis

- Incorporates LSTM (Long-Short Term Memory), and 1D Causal TCN (Temporal Convolutional Network) blocks for deep learning of sequential hand position data to BRAVE (Bravely Realtime Audio Variational autoEncoder) latent space mapping.
- Captured high FPS video and audio data to encode hand gestures for accurate positioning and inference of corresponding professional conga drum sounds.
- Applied resampling and peak detection for dataset temporal alignment.
- Used libraries including OpenCV, MediaPipe, and PyTorch.

Music-Emotion Response Study *(08/2025 - Present)*

Exploring human perceptual responses to music features in collaboration with my undergraduate honors thesis advisor.

- Composed two pop music pieces with fixed chord progressions across and variation across pitch, mode, and rhythmic speed.
- Collaborated with musicologists and data scientists on defining experimental test setup and survey metrics, to minimize confounding variables.
- Using EKG, EEG, and GSR to measure the physiological responses of study participants.

Towards Generative Adversarial Audio Synthesis via Latent Interpolation Between Desired Timbral Qualities *(05/2023)*

My senior year honors thesis exploring Generative Adversarial Networks (GANs) and multi-class classification networks.

- Designed and built a multi-class classification network to categorize sounds by their instruments with 76% accuracy.
- Researched state-of-the-art audio models including WaveGAN, WaveNet, and NSynth.
- Proposed methods to generate musical notes based on sound descriptors using a GAN.
- Wrote a peer-reviewed 24-page manuscript.

WORK EXPERIENCE

06/2023 to Current

Full Stack Software Engineer

Cisco Meraki — Remote

- Developing the Meraki Cloud Dashboard full-time on a global team of engineers.
- Technical Lead for the 'Power Settings' page of the Meraki Dashboard to distribute power-saving configuration options.
- Developed a distributed system for syncing Meraki manufacturing information across servers.
- Resolved several time-sensitive, million dollar issues for clients across the globe by debugging Meraki's 4 million line codebase.
- Performed several interviews and assisted onboarding software engineers from India and China.
- Designed an LLM for retrieving Slack and Confluence documents to help developers access information during Cisco hackathon.
- Working on RAG for an in-house test generation LLM.

01/2025 to Current

Startup CTO

Beamshyft — Boston, USA

https://beamshyft.com

- Independently designed, developed, and launched a dynamic website with video, catalogue, and scheduling system.
- Improved project clarity by outlining 4 key development phases in a statement of work.
- Adapted development phases to meet business needs.
- A B2B service for selling and shipping low-cost housing supplies directly from manufacturers to real-estate developers.

01/2023 to 12/2023

Software Engineer Intern

Cisco Meraki — Remote

- Collaborated with the UX team to develop the updated UI for the Radio Settings page using Cisco Magnetic design principles.
- Diagnosed high-priority issues to prevent dashboard outages for customers with large networks.
- Solved several mules for improved Meraki Dashboard user experience.

COURSEWORK

- **Machine Learning** (589, Graduate level)
- **Reinforcement Learning** (H589, Graduate level)
- **Artificial Intelligence** (383)
- **Uncertainty, Risk, and Decision Making** (499CM)
- **Software Entrepreneurship** (420)
- **Fundamentals of Music Theory** (110)

ACTIVITIES

- Swim-team captain
- Robotics Club (computer vision)
- St. Jude Children's Research Hospital fundraising
- TKE recruitment chairman

SKILLS

AI & ML: PyTorch, Tensorflow, MediaPipe, Multimodal VAE, RAVE, VRNN, Audio/Signal Processing, CLIP, Stable Diffusion
Programming: Python, Matplotlib, Librosa, Pandas, Embedded C, CUDA, MATLAB, React/Next.js, Ruby on Rails, SQL
Fabrication & Design: CAD (Fusion 360), PCB Design (KiCAD), 3D Printing
Creative Practice: Music Production (Ableton Live), Sound Design, Adobe
Research & Interpersonal: Problem-Solving & Critical Thinking, Creativity & Innovation (Musician, Inventor, Builder), Collaboration & Communication (Research Partnerships, Recruitment Chairman), Leadership & Initiative (Swim Team Captain, Tech Lead), Adaptability & Resilience, Industriousness

LANGUAGES

English:
Native/ Bilingual

Spanish:
Full Professional